

Лабораторная работа №6

Дисциплина: Сетевые технологии

Пакавира Арсениу Висенте Луиш

2026-03-20

1 Цель работы

- Изучить принципы распределения адресного пространства IPv4 и IPv6
- Освоить методы разбиения сетей на подсети
- Настроить и проверить работу двойного стека IPv4/IPv6
- Проанализировать изоляцию подсетей и сетевой трафик

2 Разбиение IPv4-сети на подсети (Задание 1)

Исходная сеть: 172.16.20.0/24

- Маска: 255.255.255.0
- Диапазон узлов: 172.16.20.1 – 172.16.20.254
- Broadcast: 172.16.20.255

Требуется: 3 подсети на 126 / 62 / 62 узла

3 IPv4: Результат разбиения

- **Подсеть 1:** 172.16.20.0/25
 - Узлы: 172.16.20.1 – 172.16.20.126
 - Broadcast: 172.16.20.127
- **Подсеть 2:** 172.16.20.128/26
 - Узлы: 172.16.20.129 – 172.16.20.190
 - Broadcast: 172.16.20.191
- **Подсеть 3:** 172.16.20.192/26
 - Узлы: 172.16.20.193 – 172.16.20.254
 - Broadcast: 172.16.20.255
- Подсети не пересекаются
- Адресное пространство использовано полностью

4 Разбиение IPv4-сети (Задание 2)

Исходная сеть: 10.10.1.64/26

- Маска: 255.255.255.192
- Диапазон узлов: 10.10.1.65 – 10.10.1.126

Выделение подсети на 30 узлов:

- Новый префикс: /27
- Подсеть: 10.10.1.64/27
- Диапазон узлов: 10.10.1.65 – 10.10.1.94
- Broadcast: 10.10.1.95

5 Разбиение IPv4-сети (Задание 3)

Исходная сеть: 10.10.1.0/26

- Диапазон узлов: 10.10.1.1 – 10.10.1.62

Подсеть на 14 узлов:

- Префикс: /28
- Подсеть: 10.10.1.0/28
- Узлы: 10.10.1.1 – 10.10.1.14
- Broadcast: 10.10.1.15

6 Разбиение IPv6-сети (Задание 1)

Исходная сеть: 2001:db8:c0de::/48 (документационное пространство IPv6)

- Диапазон адресов:
 - 2001:db8:c0de:: – 2001:db8:c0de:ffff:ffff:ffff:ffff:ffff

7 IPv6: Способ 1 — Subnet ID

- Используется 1 бит Subnet ID
- Новый префикс: /49

Результат:

- 2001:db8:c0de:0000::/49
- 2001:db8:c0de:8000::/49
- Деление на две равные подсети
- Interface ID остаётся 64-битным

8 IPv6: Способ 2 — Interface ID

- Выбрана подсеть /64
- Используется 1 бит Interface ID
- Новый префикс: /65

Результат:

- 2001:db8:c0de:0000::/65
- 2001:db8:c0de:0000:8000::/65
- Деление внутри одной /64
- Уменьшение Interface ID до 63 бит

9 Настройка двойного стека IPv4/IPv6 — Топология

- Создан проект в GNS3
- Используются FRR (IPv4) и VyOS (IPv6)
- Включён захват трафика

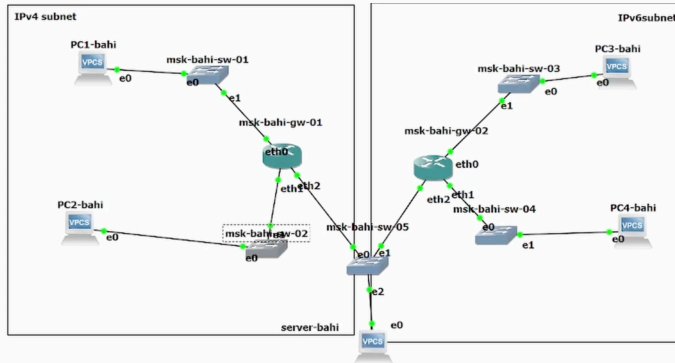
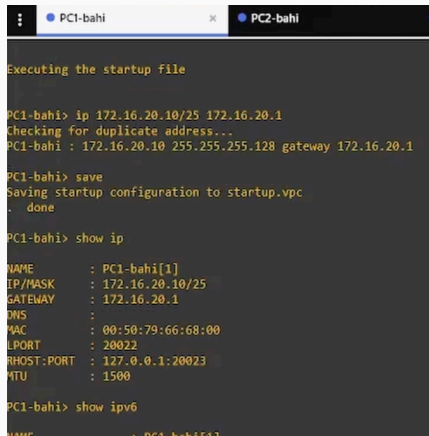


Рисунок 1: Топология сети IPv4 и IPv6

10 IPv4: Настройка PC1

- Адрес: 172.16.20.10/25
- Шлюз: 172.16.20.1
- Проверка: show ip, show ipv6



```
PC1-bahi
Executing the startup file

PC1-bahi> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
PC1-bahi : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

PC1-bahi> save
Saving startup configuration to startup.vpc
. done

PC1-bahi> show ip

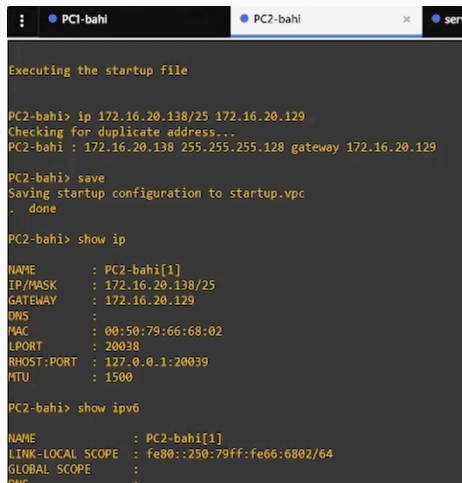
NAME       : PC1-bahi[1]
IP/MASK     : 172.16.20.10/25
GATEWAY     : 172.16.20.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20022
RHOST:PORT  : 127.0.0.1:20023
MTU         : 1500

PC1-bahi> show ipv6

NAME       : PC1-bahi[1]
```


11 IPv4: Настройка PC2

- Адрес: 172.16.20.138/25
- Шлюз: 172.16.20.129



```
PC1-bahi PC2-bahi ser

Executing the startup file

PC2-bahi> ip 172.16.20.138/25 172.16.20.129
Checking for duplicate address...
PC2-bahi : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

PC2-bahi> save
Saving startup configuration to startup.vpc
. done

PC2-bahi> show ip

NAME       : PC2-bahi[1]
IP/MASK    : 172.16.20.138/25
GATEWAY    : 172.16.20.129
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 20038
RHOST:PORT : 127.0.0.1:20039
MTU        : 1500

PC2-bahi> show ipv6

NAME       : PC2-bahi[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6802/64
GLOBAL SCOPE    :
DNS             :
```

12 IPv4: Настройка сервера

- Адрес: 64.100.1.10/24
- Шлюз: 64.100.1.1

```
VPCS> ip 64.100.1.10/24 64.100.1.1
Checking for duplicate address...
PC1 : 64.100.1.10 255.255.255.0 gateway 64.100.1.1
```

```
VPCS> save
Saving startup configuration to startup.vpc
. done
```

```
VPCS> show ip
```

```
NAME       : VPCS[1]
IP/MASK     : 64.100.1.10/24
GATEWAY     : 64.100.1.1
DNS         :
MAC         : 00:50:79:66:68:03
LPORT      : 10010
RHOST:PORT  : 127.0.0.1:10011
MTU         : 1500
```

```
VPCS> show ipv6
```

```
NAME       : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6803/64
GLOBAL SCOPE    : ::
```

13 FRR: Настройка интерфейсов

- eth0 — 172.16.20.1/25
- eth1 — 172.16.20.129/25
- eth2 — 64.100.1.1/24

```
frr# configure terminal
frr(config)# hostname msk-user-gw-01
msk-user-gw-01(config)# hostname msk-user-gw-01
msk-user-gw-01(config)# hostname msk-bahi-gw-01
msk-bahi-gw-01(config)# exit
msk-bahi-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-bahi-gw-01# configure terminal
msk-bahi-gw-01(config)# interface eth0
msk-bahi-gw-01(config-if)# ip address 172.16.20.1/25
msk-bahi-gw-01(config-if)# no shutdown
msk-bahi-gw-01(config-if)# exit
msk-bahi-gw-01(config)# interface eth1
msk-bahi-gw-01(config-if)# ip address 64.100.1.1/24
```

14 FRR: Проверка running-config

- Проверены IPv4-адреса интерфейсов
- Конфигурация сохранена

```
msk-bahi-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-bahi-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 172.16.20.1/25
exit
!
interface eth1
 ip address 64.100.1.1/24
```

15 FRR: Состояние интерфейсов

- Интерфейсы в состоянии *up*
- Адреса назначены корректно

```
msk-bahi-gw-01# show interface brief
Interface      Status  VRF      Addresses
-----
eth0           up      default  172.16.20.1/25
eth1           up      default  64.100.1.1/24
eth2           down    default
eth3           down    default
eth4           down    default
eth5           down    default
eth6           down    default
eth7           down    default
lo             up      default
pimreg         up      default
msk-bahi-gw-01#
```

Рисунок 7: FRR interface brief

16 IPv4: Проверка связности PC2 → PC1

- Использованы ping и trace
- Потерь пакетов нет

```
PC1-bahi> ping
PC1-bahi> ping 172.16.20.10/25

172.16.20.10 icmp_seq=1 ttl=64 time=0.001 ms
172.16.20.10 icmp_seq=2 ttl=64 time=0.001 ms
172.16.20.10 icmp_seq=3 ttl=64 time=0.001 ms
172.16.20.10 icmp_seq=4 ttl=64 time=0.001 ms
172.16.20.10 icmp_seq=5 ttl=64 time=0.001 ms

PC1-bahi> trace 172.16.20.10/25
tracert to 172.16.20.10, 25 hops max
 1 172.16.20.10      0.001 ms

PC1-bahi> Z
```

Рисунок 8: Ping PC2 → PC1

17 IPv4: Проверка PC2 → Server

- Связность подтверждена
- Корректный маршрут через FRR

```
PC2-bahi> ping 172.16.20.1/25  
  
host (172.16.20.129) not reachable  
  
PC2-bahi> ping 172.16.20.138/25  
  
172.16.20.138 icmp_seq=1 ttl=64 time=0.001 ms  
172.16.20.138 icmp_seq=2 ttl=64 time=0.001 ms  
172.16.20.138 icmp_seq=3 ttl=64 time=0.001 ms  
172.16.20.138 icmp_seq=4 ttl=64 time=0.001 ms  
172.16.20.138 icmp_seq=5 ttl=64 time=0.001 ms  
  
PC2-bahi> trace 172.16.20.138/25  
traceroute to 172.16.20.138, 25 hops max  
 1 172.16.20.138      0.001 ms
```

18 IPv6: Настройка сервера

- IPv6: 2001:db8:c0de:11::a/64
- Dual Stack

```
Checking for duplicate address...
PC3-bahi : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

PC3-bahi> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

PC3-bahi> save
Saving startup configuration to startup.vpc
: done

PC3-bahi> █
```

Рисунок 10: Server IPv6

19 IPv6: Настройка PC3

- IPv6: 2001:db8:c0de:12::a/64

```
PC3-bahi> ip 2001:db8:c0de:12::a/64
PC1 : 2001:db8:c0de:12::a/64

PC3-bahi> save
Saving startup configuration to startup.vpc
. done

PC3-bahi> show ip

NAME           : PC3-bahi[1]
IP/MASK        : 172.16.20.138/25
GATEWAY        : 172.16.20.129
DNS            :
MAC            : 00:50:79:66:68:01
LPORT         : 20036
RHOST:PORT     : 127.0.0.1:20037
MTU            : 1500

PC3-bahi> show ipv6

NAME           : PC3-bahi[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:db8:c0de:12::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:01
LPORT         : 20036
```

20 IPv6: Настройка PC4

- IPv6: 2001:db8:c0de:13::a/64

```
PC4-bahi> ip 2001:db8:c0de:13::a/64
PC1 : 2001:db8:c0de:13::a/64

PC4-bahi> save
Saving startup configuration to startup.vpc
. done

PC4-bahi> show ip

NAME           : PC4-bahi[1]
IP/MASK        : 0.0.0.0/0
GATEWAY        : 0.0.0.0
DNS            :
MAC            : 00:50:79:66:68:03
LPORT         : 20040
RHOST:PORT     : 127.0.0.1:20041
MTU            : 1500

PC4-bahi> show ipv6

NAME           : PC4-bahi[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6803/64
GLOBAL SCOPE    : 2001:db8:c0de:13::a/64
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:03
LPORT         : 20040
```

21 VyOS: Изменение имени

- Hostname: msk-bahis-gw-02
- Конфигурация сохранена

```
vyos@vyos:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not persist until installed
[edit]
vyos@vyos# set system host-name msk-bahi-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-bahi-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
Are you sure you want to reboot this system? [y/N] █
```

Рисунок 13: VyOS hostname

22 VyOS: IPv6 и Router Advertisement

- IPv6 назначен интерфейсам
- Включён RA

```
vyos@msk-bahi-gw-02:~$ configure
[edit]
vyos@msk-bahi-gw-02# set interfaces ethernet eth0
address 2001:db8:c0de:12::1/64
address 2001:db8:c0de:12t:1/64es ethernet eth0] already exists
address 2001:db8:c0de:12::1/64
address 2001:db8:c0de:12::1/64
address 2001:db8:c0de:12::1/64faces ethernet eth address 2001:db8:c0de:12::1/64

Invalid Ethernet interface name
Value validation failed
Set failed

[edit]
vyos@msk-bahi-gw-02#
[edit]
vyos@msk-bahi-gw-02#
[edit]
vyos@msk-bahi-gw-02#
[edit]
vyos@msk-bahi-gw-02#
[edit]
[edit]sk-bahi-gw-02# set interfaces ethernet eth0 address 2001:db8:c0de:12::1/64
de:12::/64ahi-gw-02# set service router-advert interface eth0 prefix 2001:db8:c0
[edit]
[edit]sk-bahi-gw-02# set interfaces ethernet eth1 address 2001:db8:c0de:13::1/64
de:13::/64ahi-gw-02# set service router-advert interface eth1 prefix 2001:db8:c0
```

23 VyOS: Проверка интерфейсов

- IPv6-адреса активны
- Интерфейсы *up*

```
vyos@msk-bah1-gw-02# commit

[edit]
vyos@msk-bah1-gw-02#
[edit]
vyos@msk-bah1-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bah1-gw-02# show interfaces
  ethernet eth0 {
    address 2001:db8:c0de:12::1/64
    hw-id 0c:bd:a8:04:00:00
  }
  ethernet eth1 {
    address 2001:db8:c0de:13::1/64
    hw-id 0c:bd:a8:04:00:01
  }
  ethernet eth2 {
    address 2001:db8:c0de:11::1/64
    hw-id 0c:bd:a8:04:00:02
```

24 Dual Stack Server

- Доступен из IPv4 и IPv6

```
VPCS> ping 172.16.20.10

84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=4.477 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=4.287 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=3.625 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.357 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=2.328 ms

VPCS> ping 172.16.20.138

84 bytes from 172.16.20.138 icmp_seq=1 ttl=63 time=7.726 ms
84 bytes from 172.16.20.138 icmp_seq=2 ttl=63 time=2.700 ms
84 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=3.626 ms
84 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=3.480 ms
84 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=2.780 ms

VPCS> ping 2001:db8:c0de:12::a

2001:db8:c0de:12::a icmp6_seq=1 ttl=62 time=6.117 ms
2001:db8:c0de:12::a icmp6_seq=2 ttl=62 time=9.118 ms
2001:db8:c0de:12::a icmp6_seq=3 ttl=62 time=3.035 ms
2001:db8:c0de:12::a icmp6_seq=4 ttl=62 time=3.480 ms
2001:db8:c0de:12::a icmp6_seq=5 ttl=62 time=4.632 ms

VPCS> ping 2001:db8:c0de:13::a

2001:db8:c0de:13::a icmp6_seq=1 ttl=62 time=3.363 ms
```

25 Анализ трафика: ICMP IPv4

- Echo Request / Reply
- Destination Unreachable

Wireshark packet capture showing ICMP traffic. The main pane displays a list of packets, including Echo (ping) requests and replies, and Destination Unreachable messages. Packet 49 is selected, showing its details and raw data.

No.	Time	Source	Destination	Protocol	Length	Info
27	19.361380	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x8ba9, seq=5/1280, ttl=63 (reply in 28)
28	19.361590	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x8ba9, seq=5/1280, ttl=64 (request in 27)
32	24.391294	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
34	24.396437	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
36	24.400542	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
49	410.2639...	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x12ab, seq=1/256, ttl=64 (reply in 50)
50	410.2675...	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x12ab, seq=1/256, ttl=63 (request in 49)
51	411.2704...	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x13ab, seq=2/512, ttl=64 (reply in 52)
52	411.2732...	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x13ab, seq=2/512, ttl=63 (request in 51)
53	412.2741...	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x14ab, seq=3/768, ttl=64 (reply in 54)
54	412.2772...	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x14ab, seq=3/768, ttl=63 (request in 53)
55	413.2802...	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x15ab, seq=4/1024, ttl=64 (reply in 56)
56	413.2821...	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x15ab, seq=4/1024, ttl=63 (request in 55)
57	414.2846...	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) request id=0x16ab, seq=5/1280, ttl=64 (reply in 58)
58	414.2867...	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) reply id=0x16ab, seq=5/1280, ttl=63 (request in 57)
61	422.0240...	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x1eab, seq=1/256, ttl=64 (reply in 62)
62	422.0307...	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x1eab, seq=1/256, ttl=63 (request in 61)
63	423.0323...	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x1fab, seq=2/512, ttl=64 (reply in 64)
64	423.0347...	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x1fab, seq=2/512, ttl=63 (request in 63)
65	424.0366...	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x20ab, seq=3/768, ttl=64 (reply in 66)
66	424.0397...	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x20ab, seq=3/768, ttl=63 (request in 65)
67	425.0418...	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0x21ab, seq=4/1024, ttl=64 (reply in 68)
68	425.0435...	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0x21ab, seq=4/1024, ttl=63 (request in 67)

Encapsulation type: Ethernet (1)
Arrival Time: Jan 7, 2026 21:50:58.571972000 Russia TZ 2 Standard Time
UTC Arrival Time: Jan 7, 2026 18:50:58.571972000 UTC
Epoch Arrival Time: 1767811858.571972000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 2.718000 milliseconds]
[Time delta from previous displayed frame: 6 minutes, 25.863401000 seconds]
[Time since reference or first frame: 6 minutes, 50.263943000 seconds]
Frame Number: 49
Frame Length: 98 bytes (784 bits)
Capture Length: 98 bytes (784 bits)

0000 0c 37 ba df 00 02 00 50 79 66 68
0010 00 54 ab 12 00 00 00 01 ce 0e 40
0020 14 0a 08 00 0d 60 12 ab 00 01 08
0030 0e 0f 10 11 12 13 14 15 16 17 18
0040 1e 1f 20 21 22 23 24 25 26 27 28
0050 2e 2f 30 31 32 33 34 35 36 37 38
0060 3e 3f

26 Анализ трафика: Детализация

- Ethernet + IPv4 + ICMP
- MAC-адреса и тайминги

Wireshark packet capture showing ICMP Echo (ping) requests and replies. The capture is titled "Capturing from Standard input [server-alkamal Ethernet0 to msk-alkamal-sw-05 Ethernet7]". The filter is "icmp".

No.	Time	Source	Destination	Protocol	Length	Info
3	0.003079	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x78a9, seq=1/256, ttl=63 (reply in 4)
4	0.004210	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x78a9, seq=1/256, ttl=64 (request in 3)
5	1.007512	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x79a9, seq=2/512, ttl=63 (reply in 6)
6	1.007968	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x79a9, seq=2/512, ttl=64 (request in 5)
7	2.011329	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x7aa9, seq=3/768, ttl=63 (reply in 8)
8	2.011719	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x7aa9, seq=3/768, ttl=64 (request in 7)
9	3.016686	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x7ba9, seq=4/1024, ttl=63 (reply in 10)
10	3.018669	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x7ba9, seq=4/1024, ttl=64 (request in 9)
11	4.021985	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x7ca9, seq=5/1280, ttl=63 (reply in 12)
12	4.023474	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x7ca9, seq=5/1280, ttl=64 (request in 11)
14	9.468529	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
16	9.473589	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
18	9.474729	64.100.1.10	172.16.20.10	ICMP	86	Destination unreachable (Port unreachable)
19	15.335991	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x87a9, seq=1/256, ttl=63 (reply in 20)
20	15.337185	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x87a9, seq=1/256, ttl=64 (request in 19)
21	16.342353	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x88a9, seq=2/512, ttl=63 (reply in 22)
22	16.343722	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x88a9, seq=2/512, ttl=64 (request in 21)
23	17.350563	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x89a9, seq=3/768, ttl=63 (reply in 24)
24	17.350885	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x89a9, seq=3/768, ttl=64 (request in 23)
25	18.356154	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x8aa9, seq=4/1024, ttl=63 (reply in 26)
26	18.356786	64.100.1.10	172.16.20.10	ICMP	98	Echo (ping) reply id=0x8aa9, seq=4/1024, ttl=64 (request in 25)
27	19.361380	172.16.20.10	64.100.1.10	ICMP	98	Echo (ping) request id=0x8ba9, seq=5/1280, ttl=63 (reply in 28)

Packet 14 details:

```
[Time delta from previous captured frame: 853.000 microseconds]
[Time since reference or first frame: 3.079000 milliseconds]
Frame Number: 3
Frame Length: 98 bytes (784 bits)
Capture Length: 98 bytes (784 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
```


27 Самостоятельно: Подсеть 10.10.1.96/27

- 30 хостов
- Маска /27

IP калькулятор

IP адрес:

Маска:

Подсчитать

	Значение	Бинарное значение
Address:	10.10.1.96	00001010.00001010.00000001.011 100000
Bitmask:	27	
Netmask:	255.255.255.224 = 27	11111111.11111111.11111111.111 00000
Wildcard:	0.0.0.31	00000000.00000000.00000000.000 11111
Network:	10.10.1.96	00001010.00001010.00000001.011 00000 (Class A)
Broadcast:	10.10.1.127	00001010.00001010.00000001.011 11111

28 Самостоятельно: Подсеть 10.10.1.16/28

- 14 хостов
- Маска /28

IP адрес:

10.10.1.16

Маска:

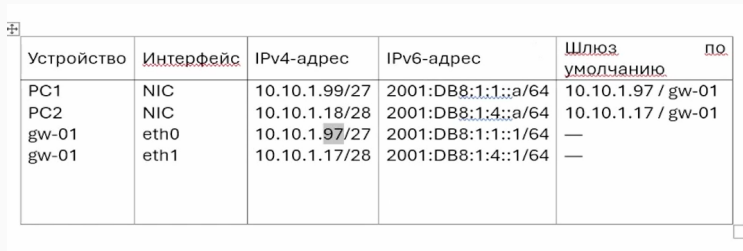
28 - 255.255.255.252 ▾

Подсчитать

	Значение	Бинарное значение
Address:	10.10.1.16	00001010.00001010.00000001.0001 0000
Bitmask:	28	
Netmask:	255.255.255.240 = 28	11111111.11111111.11111111.1111 0000
Wildcard:	0.0.0.15	00000000.00000000.00000000.0000 1111
Network:	10.10.1.16	00001010.00001010.00000001.0001 0000 (Class A)
Broadcast:	10.10.1.31	00001010.00001010.00000001.0001 1111
HostMin:	10.10.1.17	00001010.00001010.00000001.0001 0001

29 Таблица адресации

- Минимальные адреса на router
- Корректная схема



Устройство	Интерфейс	IPv4-адрес	IPv6-адрес	Шлюз по умолчанию
PC1	NIC	10.10.1.99/27	2001:DB8:1:1::a/64	10.10.1.97 / gw-01
PC2	NIC	10.10.1.18/28	2001:DB8:1:4::a/64	10.10.1.17 / gw-01
gw-01	eth0	10.10.1.97/27	2001:DB8:1:1::1/64	—
gw-01	eth1	10.10.1.17/28	2001:DB8:1:4::1/64	—

Рисунок 21: Addressing table

30 GNS3: Переименование устройств

- Имена по шаблону
- Топология сохранена

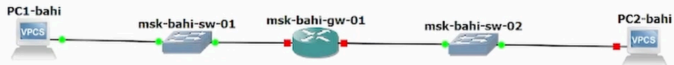


Рисунок 22: GNS3 names

31 PC1-bahis: IPv4/IPv6

- IPv4 + IPv6 назначены

```
PC1-bahi> ip 10.10.1.96/27
Invalid host address

PC1-bahi> ip 10.10.1.96/27
Invalid host address

PC1-bahi> ip 10.10.1.97/27
Checking for duplicate address...
PC1-bahi : 10.10.1.97 255.255.255.224

PC1-bahi> save
Saving startup configuration to startup.vpc
. done

PC1-bahi> show ip

NAME       : PC1-bahi[1]
IP/MASK     : 10.10.1.97/27
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:00
I.P.O.R.T  : 20000
```

32 PC1-bahis: Проверка

- show ip
- show ipv6

```
PC1-bahi> show ipv6

NAME                : PC1-bahi[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        :
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:00
LPORT                : 20008
RHOST:PORT           : 127.0.0.1:20009
MTU                  : 1500
```

Рисунок 24: PC1 verify

33 PC2-bahis: IPv4/IPv6

- IPv4 + IPv6 назначены

```
PC2-bahi> ip 10.10.1.17/28
Checking for duplicate address...
PC2-bahi : 10.10.1.17 255.255.255.240

PC2-bahi> save
Saving startup configuration to startup.vpc
. done

PC2-bahi> show ip

NAME       : PC2-bahi[1]
IP/MASK     : 10.10.1.17/28
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20010
RHOST:PORT  : 127.0.0.1:20011
MTU         : 1500
```

34 PC2-bahis: Проверка

- Параметры корректны

```
PC2-bahi> show ipv6  
  
NAME                : PC2-bahi[1]  
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64  
GLOBAL SCOPE        :  
DNS                 :  
ROUTER LINK-LAYER   :  
MAC                 : 00:50:79:66:68:01  
LPORT               : 20010  
RHOST:PORT           : 127.0.0.1:20011  
MTU                  : 1500  
  
PC2-bahi> █
```

Рисунок 26: PC2 verify

35 VyOS: Интерфейсы (IPv4/IPv6)

- Минимальные адреса
- RA активен

```
vyos@vyos:~$ configure
WARNING: You are currently configuring a live-ISO environment, changes will not persist until installed
edit]
vyos@vyos# set interfaces ethernet eth0 address 2001:db8:1:1::1/64
edit]
vyos@vyos# set service router-advert interface eth0 prefix 2001:db8:1:1::/64
edit]
vyos@vyos# set interfaces ethernet eth1 address 2001:db8:1:4::1/64
edit]
vyos@vyos# set service router-advert interface eth0 prefix 2001:db8:1:4::/64
edit]
vyos@vyos# compare
edit interfaces ethernet eth0]
address 2001:db8:1:1::1/64
edit interfaces ethernet eth1]
address 2001:db8:1:4::1/64
edit]
service {
  router-advert {
    interface eth0 {
      prefix 2001:db8:1:1::/64 {
      }
      prefix 2001:db8:1:4::/64 {
      }
    }
  }
}
edit]
vyos@vyos# commit
```

36 VyOS: Проверка

- show interface

```
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 2001:db8:1:1::1/64
    hw-id 0c:37:d8:69:00:00
  }
  ethernet eth1 {
    address 2001:db8:1:4::1/64
    hw-id 0c:37:d8:69:00:01
  }
  ethernet eth2 {
    hw-id 0c:37:d8:69:00:02
  }
  loopback lo {
```

- Освоены методы разбиения IPv4 и IPv6 сетей
- Реализована корректная адресация и маршрутизация
- Настроен и проверен Dual Stack
- Подтверждена изоляция IPv4 и IPv6
- Анализ ICMP-трафика подтвердил корректность конфигурации